Steady State Cornering Case: Tire Forces

Finding Lateral Acceleration:



r = Turning radius

$a_c = \frac{v^2}{r} = \mu g$

$G_{Axial} = \frac{a_y}{g} = \frac{v^2}{rg}$

$a_{cmax} = \mu g \longrightarrow \frac{a_{cmax}}{g} = \mu$ = later force in Gs = 0.75

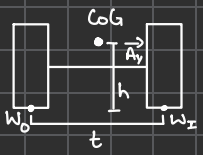
Lateral Load Transfer:

$\Delta W = \frac{WA_y h}{t} = \frac{(2670)0.75(0.74631)}{1.1335} = 1318.470026$

- On Front Outside tire:

Assume front axle recieves 40% of total lateral load transfer

$0.40(1318.47) = \underline{527.3880106\ N}$



Vertical load on Outside Tire:

40/60 Weight distribution

$F_{out} = \frac{(0.4)(2670)}{2} + 527.388 = \underline{1061.38801\ N}$

Axial load on Tire:

$W_t = G \cdot W_r = (0.75)(1061.38801)$

$= \underline{796.0410083\ N}$

Calculating Bearing Loads:

$R_s = 0.2921\ m$, $a = b = 0.018\ m$, $l = 0.036\ m$,

- Radial Load:

$F_{ro} = \left| \frac{bW_r - R_s W_t}{l} \right|$

$= \left| \frac{(0.018)(1061.39) - (0.2921)(796.04)}{0.036} \right|$

$= \underline{5928.293225\ N}$

$F_{ri} = \left| \frac{aW_r + R_s W_t}{l} \right|$

$= \left| \frac{(0.018)(1061.39) + (0.2921)(796.04)}{0.036} \right|$

$= \underline{6989.675111\ N}$

Service life Requirements:

$L_{10} = \frac{10^6}{60n} L_{10}$ 51 hrs needed

Wheel speed = $n = \frac{60v}{2\pi R_s} = \frac{60(11.176)}{2\pi(0.2921)}$

$= 365.36$ rpm

Total revs = $N = n(60)t = 1.118$ million revs

$\underline{L_{10} \geq 51\ hrs}$

Bow level axial loading and life verification pending SKF internal-load analysis